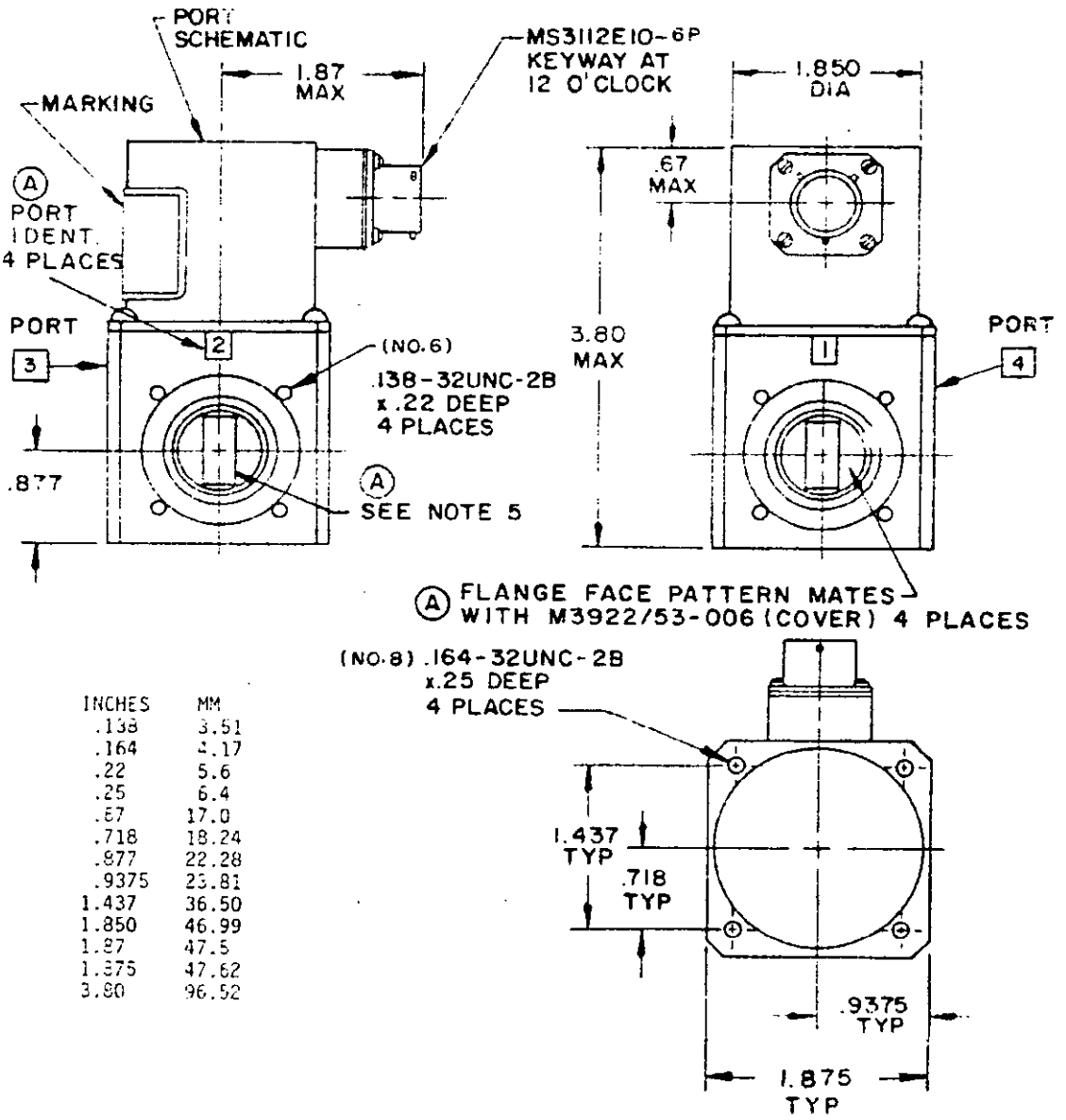
MILITARY SPECIFICATION SHEET

SWITCHES, WAVEGUIDE, 2P2T, FAIL-SAFE

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for acquiring the switches described herein shall consist of this specification and the latest issue of MIL-S-55041.



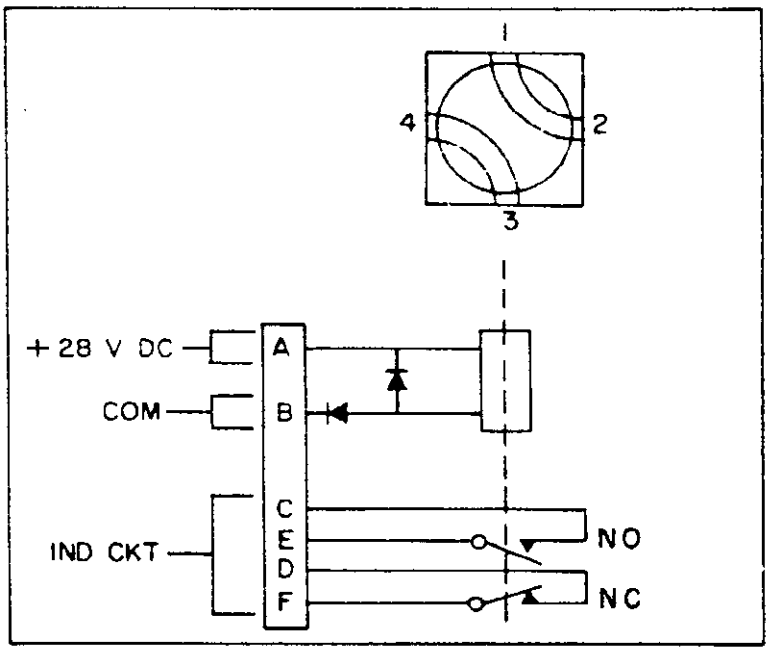
INCHES	MM
.138	3.51
.164	4.17
.22	5.6
.25	6.4
.67	17.0
.718	18.24
.877	22.28
.9375	23.81
1.437	36.50
1.850	46.99
1.87	47.5
1.875	47.62
3.80	96.52

FIGURE 1. Switch configuration and schematic, part no. M55041/7-001.

Ⓐ denotes changes

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Check the source to verify that this is the current version before use.

SCHEMATIC

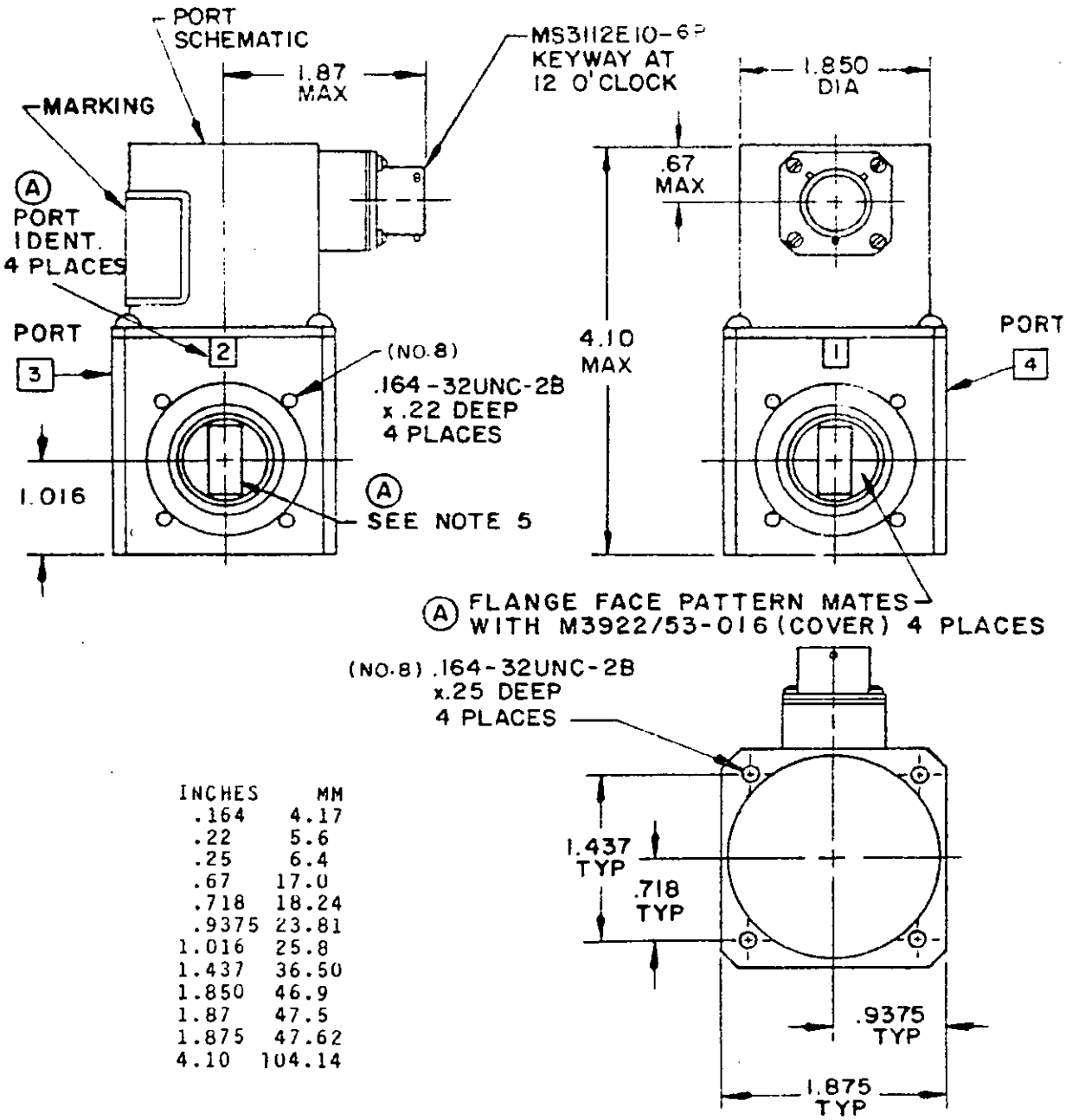


SHOWN IN DEENERGIZED POSITION

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only and are based upon 1.00 inch = 25.4 mm.
3. Round corners of case may be squared.
4. Unless otherwise specified, tolerances are ±.010 (.25 mm) for three place decimals and ±.03 (.8mm) for two place decimals.
5. Waveguide size equivalent to MIL-W-85/1-091.
6. DC polarity must be observed.

Ⓐ FIGURE 1. Switch configuration and schematic, part no. M55041/7-001 - Continued.

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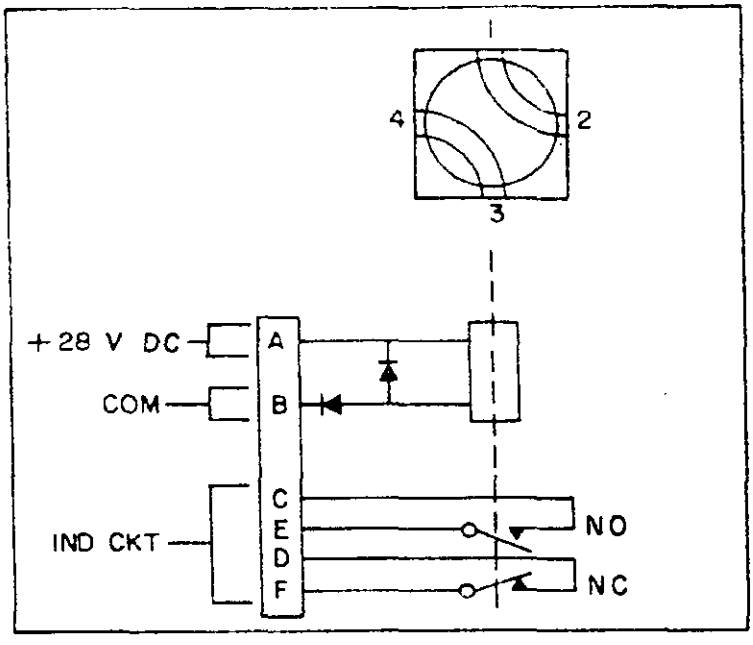


INCHES	MM
.164	4.17
.22	5.6
.25	6.4
.67	17.0
.718	18.24
.9375	23.81
1.016	25.8
1.437	36.50
1.850	46.9
1.87	47.5
1.875	47.62
4.10	104.14

FIGURE 2. Switch configuration and schematic, part no. M55041/7-....

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SCHEMATIC

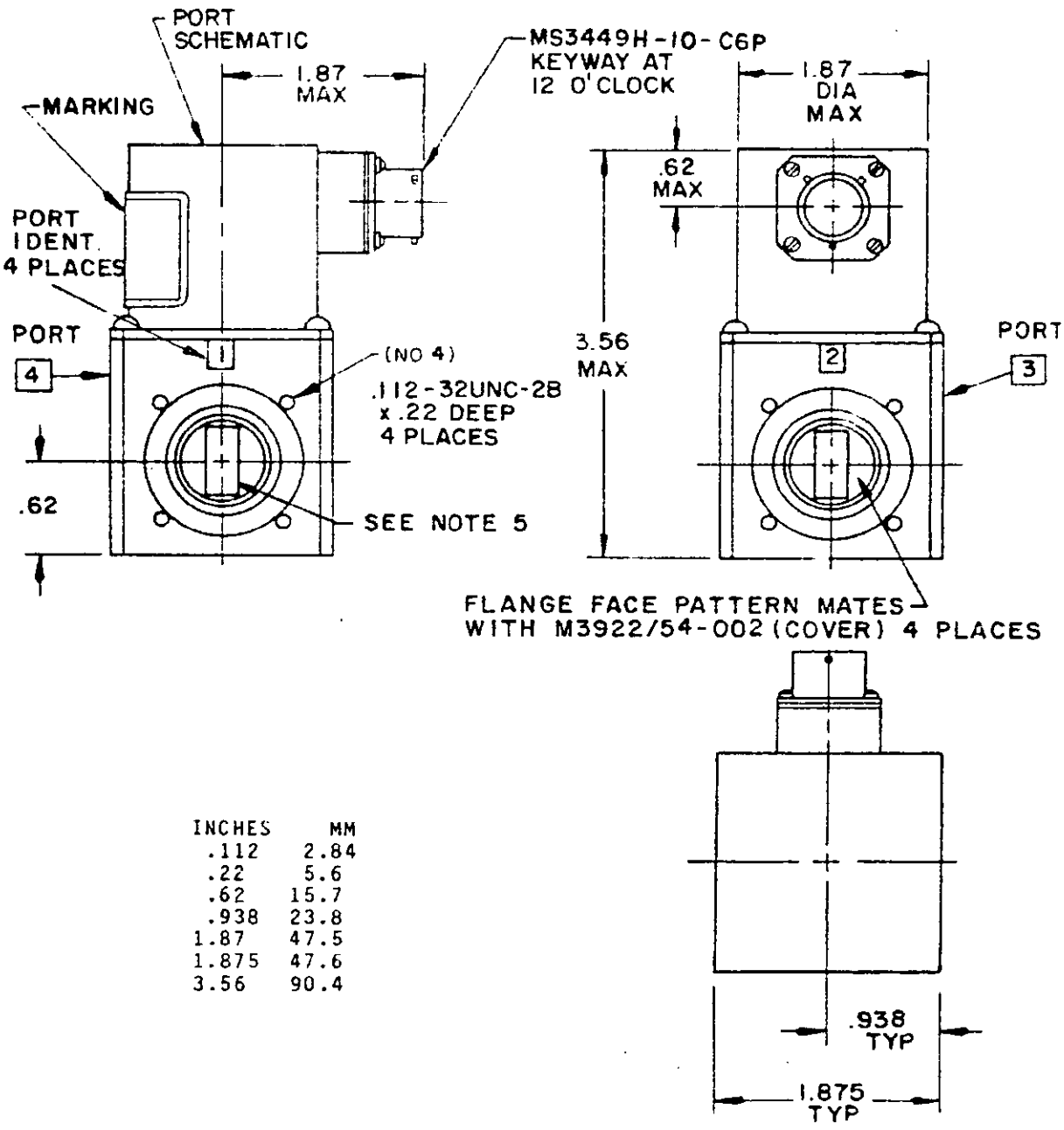


SHOWN IN DEENERGIZED POSITION

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only and are based upon 1.00 inch = 25.4 mm.
3. Round corners of case may be squared.
4. Unless otherwise specified, tolerances are ±.010 (.25 mm) for three place decimals and ±.03 (.8 mm) for two place decimals.
5. Waveguide size equivalent to MIL-W-85/1-078.
6. DC polarity must be observed.

Ⓐ FIGURE 2. Switch configuration and schematic, part no. M55041/7-002 - Continued.

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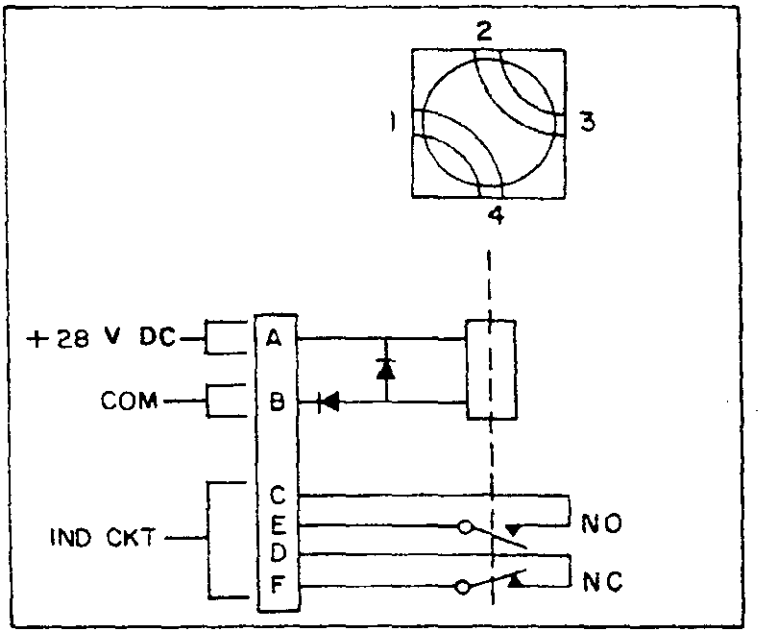


INCHES	MM
.112	2.84
.22	5.6
.62	15.7
.938	23.8
1.87	47.5
1.875	47.6
3.56	90.4

Ⓐ FIGURE 3. Switch configuration and schematic, part no. M55041/7-003.

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SCHEMATIC

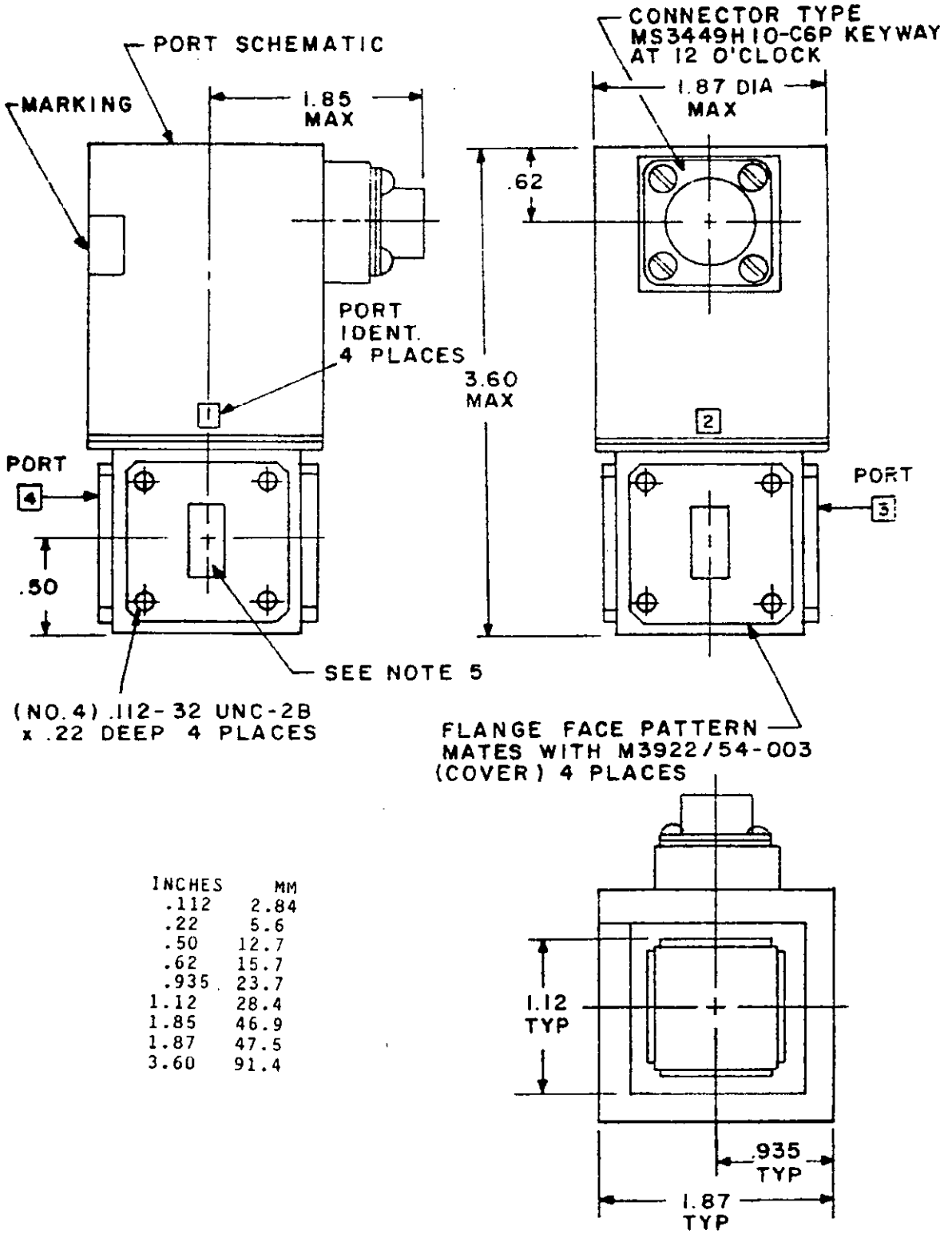


SHOWN IN DEENERGIZED POSITION

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only and are based upon 1.00 inch = 25.4 mm.
3. Round corners of case may be squared.
4. Unless otherwise specified, tolerances are ±.010 (.25 mm) for three place decimals and ±.03 (.8 mm) for two place decimals.
5. Waveguide size equivalent to MIL-W-85/1-103.
6. DC polarity must be observed.

Ⓐ FIGURE 3. Switch configuration and schematic, part no. M55041/7-003 - Continued.

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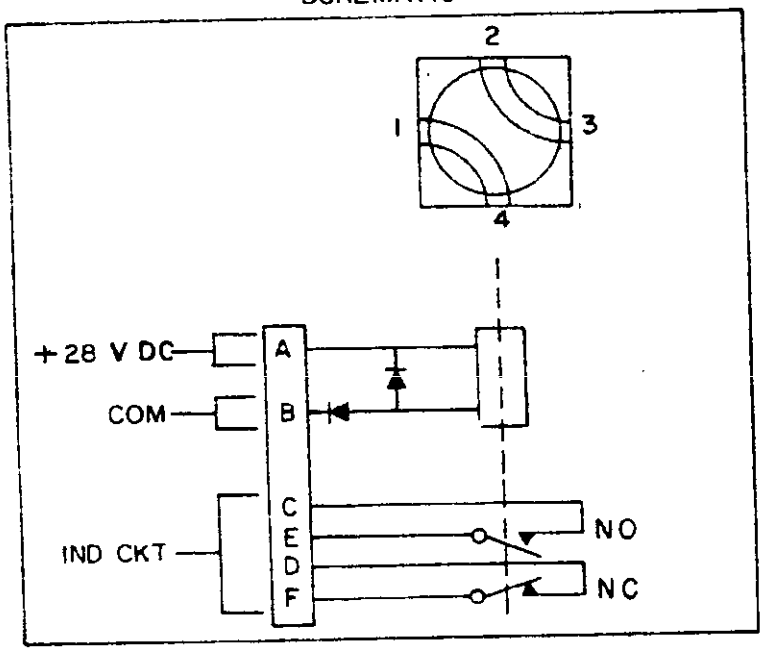


INCHES	MM
.112	2.84
.22	5.6
.50	12.7
.62	15.7
.935	23.7
1.12	28.4
1.85	46.9
1.87	47.5
3.60	91.4

Ⓐ FIGURE 4. Switch configuration and schematic, part no. M55041/7-004

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SCHEMATIC



SHOWN IN DEENERGIZED POSITION

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only and are based upon 1.00 inch = 25.4 mm.
3. Round corners of case may be squared.
4. Unless otherwise specified, tolerances are ±.010 (.25 mm) for three place decimals and ±.03 (.8 mm) for two place decimals.
5. Waveguide size equivalent to MIL-W-85/3-006.
6. DC polarity must be observed.

Ⓐ FIGURE 4. Switch configuration and schematic, part no. M55041/7-004 - Continued.

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Ⓐ TABLE I. Electrical performance characteristics.

Dash number	Figure number	Housing	Operating frequency range	Position indicator circuit	Interlock circuit	Nominal operating voltage	Pull-in voltage (less than)	Dropout voltage (less than)	Operating current (max) 3/	Holding current (max)
			GHz	3/ amperes		V dc	V dc	V dc	amperes	amperes
001	1	1/ 2/	12.4-18.0	2	N/A	28	20	10	0.5	0.5
002	2	1/ 2/	8.2-12.4	3	N/A	28	20	10	0.5	0.5
003	3	1/ 2/	18-26.0	3	N/A	28	20	10	1.2	0.5
004	4	1/ 2/	26-40.0	3	N/A	28	20	10	1.2	0.5

Ⓐ TABLE I. Electrical performance characteristics - Continued.

Dash number	RF power handling capability	Isolation (min)	VSWR (max)	Insertion loss (max)	Switching time	Waveguide pressurization	Life cycles (X1000)	Weight (max)
	(Avg)	dB		dB	ms	lb_f/in^2		oz
001	1000 watts	65	1.10:1	0.1	100	20	200	21
002	1062 watts	70	1.10:1	0.1	100	20	200	23
003	42 kW (peak) 4/	50	1.10:1	0.1	100	45	100	19
004	22 kW (peak) 4/	50	1.10:1	0.1	100	45	100	19

1/ Enclosed, sand- and dust-proof when waveguide flanges are capped.

2/ Rotor and housing of aluminum alloy.

3/ At 28 V dc.

4/ Duty cycle of 1×10^{-3}.

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REQUIREMENTS:

Design and construction:

Dimensions and configuration: See figures.

Power connectors: See figures.

Operation: Break before make.

Operating temperature range: -54°C to +95°C.

Electrical performance characteristics: See table I.

Part number: M55041/7- (and dash number from table I).

(A) Substitutability data: The switches covered by this specification sheet are substitutable for the user's or manufacturer's part numbers specified in table II. This information in no way implies that the manufacturer's part number or NSN is suitable as a substitute for the military part number.

(A) TABLE II. Substitutability data.

Dash number	Manufacturer's code	Commercial type	NSN 5985-
001	97384	5339-096016	---
001	82152	33D00100-10	00-163-2286
002	82152	33D01100	01-026-6486
003	33472	112863-01	---
003	01456	42SR16-4-28	---
004	33472	112864-01	---
004	01456	28SR16-4-28	---

Custodians:
Army - ER
Navy - EC
Air Force - 85

Review activities:
Army - MI, SM
Air Force - 11, 17, 99
DLA - ES

User activities:
Navy - AS, MC, OS, SH
Air Force - 19

Agent:
DLA - ES

Preparing activity:
Navy - EC

(Project 5985-0975-4)

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